

A Review of Vision-Language-Action Models and Large Language Models for Autonomous Driving

A Study Based on Recent Research (2022–2026)

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Abstract

By the use of Artificial Intelligence like Large Language Models(LLMs),Vision Language action(VLA), the autonomous driving has rapidly evolved.The earlier or older vehicle systems depends on the modular perception, planning and control, which frequently have problem to understand the complex situations, have problem to adjust with new situations, and also have problem in communicating with driver effectively.New Ai systems has evolved much into “all in one” systems that can process every information by its own.like information of sighting,sounds,as easily as text.these models can do different task by there own in same program instead of using separate tools.This review studies the recent papers from 2022-2026 based on LLMs and VLA systems for automatic Driving,the papers include the following models, V2V-LLM, Max-V1, LVLDrive, AutoMoT and related models.This review examines the following parameters like technical architectures, datasets, benchmark performance, key innovations, and emerging challenges. Major trends include multimodal fusion, spatial reasoning, cooperative vehicle intelligence, prompt-based driving policies, and end-to-end action generation.While having great progress, there are some problems also related to safety,interpreting,efficiency of computation,and real world deployment.suggestions for futures are to create lightweight VLA architectures, neuro-symbolic reasoning, and safer human-centered autonomous systems.

Keywords

Autonomous Driving, Large Language Models, Vision-Language Models, Vision-Language-Action Models, Multimodal AI

Introduction

Autonomous Driving is one of the most challenging applications of Artificial intelligence,which requires strong and reliable senses,reasoning, decision making, and great control in highly dynamic environment. Old-school self-driving systems work like a relay team, passing information from one specialized department to another—sensing the road, predicting movement, then planning and steering. While this works, it’s a bit rigid.but if one system makes a mistake, then it will make other systems to make mistakes which will lead to bigger errors, which we cant afford.

Recent Developments in foundation models like Large Language Models(LLMs) and Vision Language Models(VLMs) have opened the new chances for more intelligent automatic systems. As, these systems combines visual trust, language reasoning, and generation of action, which leads to the development of Vision Language Action(VLA) systems.

VLA systems promise:

- End-to-end driving policies
- Improved spatial reasoning
- Human-like instruction following
- Better human-vehicle interaction
- Cooperative multi-agent driving intelligence

This review examines recent advancements in LLM and VLA-based autonomous driving systems, compares methodologies and results, identifies challenges, and discusses future directions.

Review Methodology

This review is based on selected research papers published from 2022 to 2026 from:

- arXiv
- IEEE Xplore
- Scopus
- Google Scholar
- DBLP

Selection Criteria

Included papers satisfy:

- Focus on autonomous driving
- Use of LLMs, VLMs or VLA models
- Benchmark or dataset evaluation
- Novel framework or survey contribution

Reviewed Papers

Major works examined include:

- Jiang et al. (VLA Survey)
- Chiu et al. (V2V-LLM)
- Yang et al. (Max-V1)
- Wei et al. (LVLDriver)
- Wu et al. (EnhanceVLM)
- Acharya (LLM Integration)
- Zhao et al. (VLM-driven HVI)
- Elhenawy et al. (CLIP-based VLM)
- Huang et al. (AutoMoT)
- Chowdhery et al. (PaLM)

Literature Review

3.1 Evolution of LLMs and VLA Models in Autonomous Driving

Large language models like PaLM shows that scaling leads to strong reasoning and generalization abilities. Although originally designed for NLP, these models motivated their extension into multimodal autonomous systems.

Key contribution:

Large-scale reasoning foundation

Transformer scalability

Instruction-following capability

3.2 Vision-Language Models for Driving

VLMs combine vision with language for scene understanding and decision support.

CLIP-Based Dynamic Scene Understanding

Elhenawy et al. fine-tuned CLIP for autonomous driving scene understanding.

Results:

- F1 score: 91.1%
- Outperformed GPT-4o in complex scenarios
- Real-time inference capability

Contribution:

- Strong ADAS scene reasoning
- Efficient deployment

3.3 Prompt-Based Spatial Reasoning

The **EnhanceVLM** system uses specific "instructions" (prompts) to help the AI understand its physical surroundings better.

Features:

- Mixture-of-Prompts routing
- Chain-of-thought prompting
- Visual reasoning modules

Performance:

- 70.87% Phase-1 accuracy
- 72.85% Phase-2 accuracy

→ demonstrates prompting as a strong improvement strategy for VLM driving systems.

4. Vision-Language-Action Models

4.1 VLA Survey Framework

Jiang et al. provided a taxonomy of VLA systems integrating:

- Perception
- Language reasoning
- Action generation

The study consolidated:

- 20+ VLA models
- Benchmarks:
 - nuScenes
 - CARLA
 - Bench2Drive

→ It Finds VLA models as “action gap” between reasoning and control reducer.

4.2 Max-V1

Max-V1 proposed a lightweight end-to-end VLM for trajectory planning.

Architecture:

- Single-pass vision-language model
- Autoregressive waypoint prediction

Performance:

- Average L2 error: 0.21 m
- Max error: 0.30 m
- 30% improvement over prior models

Advantages:

- Lean architecture
- Strong zero-shot generalization

4.3 AutoMoT

AutoMoT uses a system where different AI components **work on their own time** rather than waiting to stay in sync.

Key idea:

- Fast-slow asynchronous inference
- Shared attention for reasoning and control

Contributions:

- Reduced latency
- Competitive with state-of-the-art end-to-end models

→ This addresses computational inefficiency in unified VLA systems.

5. Cooperative and Multimodal Driving Intelligence

5.1 V2V-LLM

Chiu et al. proposed cooperative driving using multimodal LLMs.

Inputs:

- LiDAR
- Vision
- Language QA

Dataset:

- V2V-QA (~48K frames)

Results:

- L2 error reduced to 4.99m
- Better than 6.72m baseline
- Improved collision avoidance

Contribution:

→ Vehicle-to-vehicle cooperation moves beyond perception into planning.

5.2 LVLDrive

LVLDrive addresses weak spatial reasoning in image-only VLMs.

Method:

- LiDAR + VLM fusion
- Gradual fusion Q-former
- 3D spatial reasoning

Outcome:

- Superior to vision-only VLMs
- Improved scene understanding and decision safety

6. Human-Vehicle Interaction with LLMs

Human-centered autonomous systems represent another growing research direction.

Acharya (2025)

Integrated:

- LLM reasoning
- Multi-head fusion
- LoRA and RLHF adaptations

Results:

- 16% reduction in perception errors
- 24% lower disengagement
- 34% increase in trust

Zhao et al. (2026)

Focused on pedestrian-vehicle interaction.

Results:

- +2.8% mAP over YOLOX
- Better occlusion handling
- Safety improvements in CARLA

→These studies show LLMs improving interaction, trust and interpretability.

Comparative Analysis

Model	Major Contribution	Dataset	Key Result
V2V-LLM	Cooperative driving	V2V-QA	L2 error 4.99m
Max-V1	End-to-end planning	nuScenes	30% improvement
LVLDrive	Spatial reasoning	SA-QA	Outperformed vision-only models
EnhanceVLM	Prompt reasoning	RoboSense	72.85% accuracy
CLIP-VLM	Scene understanding	Honda Scenes	91.1% F1
AutoMoT	Unified VLA model	Multiple	SOTA-level results

Emerging Trends

From reviewed Papers, major trends include:

1 Multimodal Fusion

Combining:

- Vision
- LiDAR
- Language
- Trajectories

is becoming standard.

2 End-to-End Driving Policies

Shift from modular pipelines to unified VLA policies.

Examples:

- Max-V1
- AutoMoT

3 Spatial Reasoning

Putting more focus on 3D shapes and layout

Examples:

- LVLDrive
- EnhanceVLM

4 Cooperative Intelligence

Vehicle-to-vehicle multimodal reasoning is emerging.

Example:

- V2V-LLM

Research Challenges

While Having Progress, There Are Major Challenges Remaining

Safety

Problem of hallucination in LLM-Driven control

Interpretability

Decision Making mostly remains blur

Computational Cost

Many models require:

- High GPU resources
- Large memory
- High latency

Real-world Deployment

It is tough to take what an AI learns in a perfect video game and make it work safely on messy, unpredictable real-world streets.

Benchmark standardization

Right now, there's no single, agreed-upon "final exam" to test these systems. Because every researcher uses different rules and scoring, it's hard to fairly compare which AI is actually the safest or smartest.

In even simpler terms: Everyone is grading their own homework using different scales, so we don't have a universal way to pick a winner.

Future Directions

Lightweight VLA Models

Develop efficient, compressed architectures that enable low-latency, real-time Vision-Language-Action processing on edge devices with limited computational resources and power.

Neuro-Symbolic Reasoning

Integrate deterministic logical constraints with probabilistic LLM reasoning to ensure verifiable safety, robust causal inference, and adherence to traffic laws.

Safer Human-Centered Driving AI

Focus on transparent decision-making and natural language explanations to foster user trust and improve the seamless interaction between humans and autonomous systems.

Real-Time Cooperative Autonomous Intelligence

Advance multi-agent communication protocols that allow large-scale fleets to synchronize movements, share environmental data, and optimize global traffic flow collectively.

Standardized VLA Benchmarks

Establish universal evaluation frameworks and diverse, high-fidelity datasets to enable objective performance comparisons across different Vision-Language-Action driving architectures and models.

Conclusion

Large language and vision-language-action models are transforming autonomous driving from modular reactive systems into reasoning-driven intelligent agents. Recent studies demonstrate substantial advances in multimodal fusion, spatial understanding, cooperative driving, and human-vehicle interaction. Models such as V2V-LLM, Max-V1, LVLDriVe and AutoMoT show the potential of unified perception-to-action systems. However, safety, interpretability, computational efficiency, and deployment remain open problems. Continued progress in lightweight VLA architectures, neuro-symbolic reasoning, and benchmark development will be critical for the future of reliable autonomous vehicles.

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